

# Empirical and Molecular Formulas

Simplest ratios, formulas from percentages, and molecular formulas

Name \_\_\_\_\_ Class \_\_\_\_\_ Date \_\_\_\_\_

## DATA

H = 1.008   C = 12.01   N = 14.01   O = 16.00   Ca = 40.08   Fe = 55.85

Show your working: % → g → mol → divide by the smallest → whole numbers.

## A. Understanding

1. What is the difference between an empirical formula and a molecular formula?

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2. Write the empirical formula of: (a)  $C_6H_6$ , (b)  $C_2H_4$ , (c)  $H_2O_2$ , (d)  $C_8H_{18}$ , (e)  $N_2O_4$ .

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3. After dividing by the smallest number of moles, a student gets a ratio of 1 : 1.50 and writes the formula as  $AB_2$ . What is wrong, and what is the correct ratio?

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## B. Empirical formulas from percent composition

4. A hydrocarbon is 85.63% carbon and 14.37% hydrogen by mass. Find its empirical formula.

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5. A compound is 40.04% calcium, 12.00% carbon and 47.96% oxygen by mass. Find its empirical formula.

6. An oxide of iron is 72.36% iron and 27.64% oxygen by mass. Find its empirical formula.

### C. Molecular formulas

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7. A compound has empirical formula  $\text{CH}_2\text{O}$  and molar mass 60.05 g/mol. Find its molecular formula.

8. Acetylene has empirical formula  $\text{CH}$  and molar mass 26.04 g/mol. Find its molecular formula.

### D. Challenge

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9. A compound of nitrogen and oxygen is 30.43% nitrogen by mass, and its molar mass is 92.02 g/mol. Find its empirical and molecular formulas.